

RuFLEX: A Hybrid Deep Fuzzy Learning Platform for Geodata with Interpretable Analysis of Rules, Factors, and Hidden Concepts

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Abstract

This paper presents RuFLEX, a platform for hybrid deep fuzzy learning oriented toward geanalytics, spatially informed classification, forecasting, and intelligent decision support. The platform combines interpretable rule-based mechanisms, differentiable fuzzy layers, and deep fuzzy feature-learning architectures within a unified workflow implemented in Python and PyTorch. Experimental validation was carried out on California Housing Regression and California Value Binary Geo.

Keywords: deep fuzzy learning; hybrid neuro-fuzzy models; deep fuzzy feature learning; explainability; geanalytics; interpretable machine learning

I. Introduction

RuFLEX is designed as a platform rather than a single fixed architecture. It extends a vendored deep fuzzy backend with its own SDK, project model, explainability tooling, visual workbench, and reproducible experiment layer. This separation enables both backend continuity and independent platform evolution.

The main objective is to support multiple hybrid deep fuzzy modeling modes while preserving interpretable reasoning at the level of membership functions, rules, hidden concepts, and factor contributions.

II. Platform Architecture and Modeling Modes

The current architecture includes a core layer with variables, terms, membership functions, rules, and model specifications; a model layer with flat neuro-fuzzy baselines and deep fuzzy feature learning; a training layer with bootstrap

initialization, stage-wise pretraining, refinement, and fine-tuning; and an explainability layer with rule records, concept flow, and dashboard payloads.

- Python SDK and toolbox-style API
- visual workbench
- differentiable fuzzy training contour
- Explainability Dashboard
- reproducible article benchmark pipeline

III. Experimental Setup

Two geo-oriented tabular tasks were used in validation: California Housing Regression and California Value Binary Geo. The first task preserves the spatial predictors Latitude and Longitude, while the second task uses the same source data to define a balanced binary classification problem based on housing value.

The compared configurations were Flat Baseline, Flat Interpretable, Deep Article Demo, and Deep Research.

IV. Results and Discussion

On California Housing Regression, the best configuration was Deep Research with ``test_rmse = 0.795757`` and ``test_r2 = 0.507159``. On California Value Binary Geo, the best configuration was Flat Interpretable with ``test_accuracy = 0.746341`` and ``test_f1 = 0.739348``. This supports the platform-oriented design: different fuzzy regimes may become preferable depending on task structure.

Table I. Regression benchmark results

Configuration	Test RMSE	Test MAE	Test R2
Flat Baseline	0.890950	0.678868	0.382192
Flat Interpretable	1.115952	0.880786	0.030747
Deep Article Demo	0.816314	0.610628	0.481367
Deep Research	0.795757	0.593770	0.507159

Table II. Classification benchmark results

Configuration	Test Accuracy	Test Precision	Test F1
Flat Baseline	0.742683	0.739899	0.735257
Flat Interpretable	0.746341	0.743073	0.739348

Deep Article Demo	0.739024	0.724221	0.738386
Deep Research	0.743902	0.741772	0.736181

V. Explainability Figures

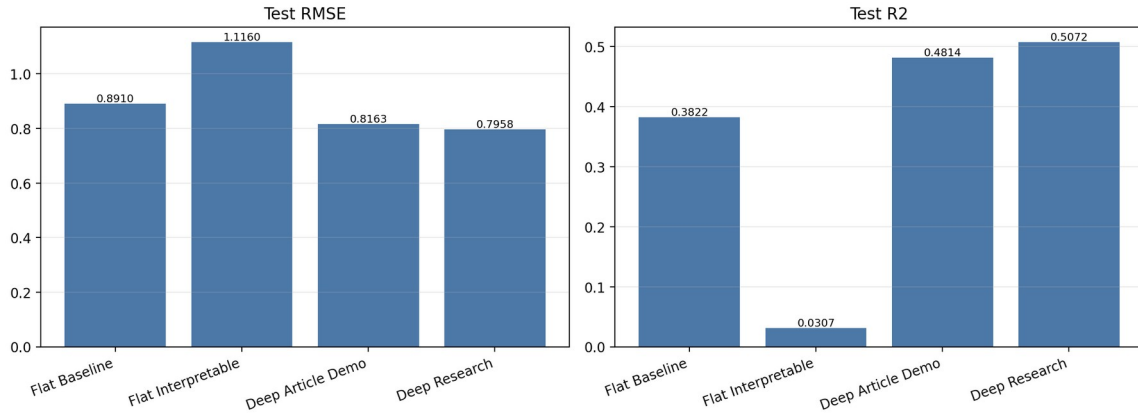


Fig. 1. Comparison of flat and deep fuzzy configurations on California Housing Regression.



Fig. 2. Explainability board for the best regression configuration.

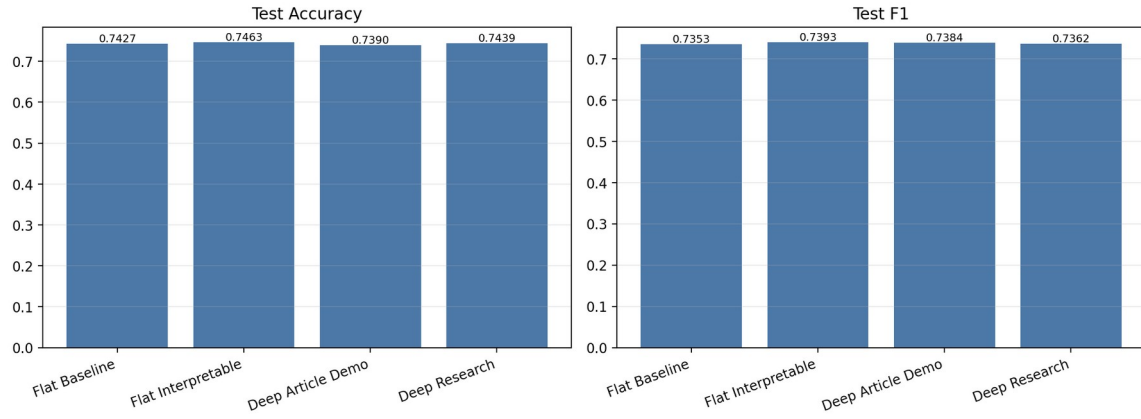


Fig. 3. Comparison of RuFLEX configurations on California Value Binary Geo.

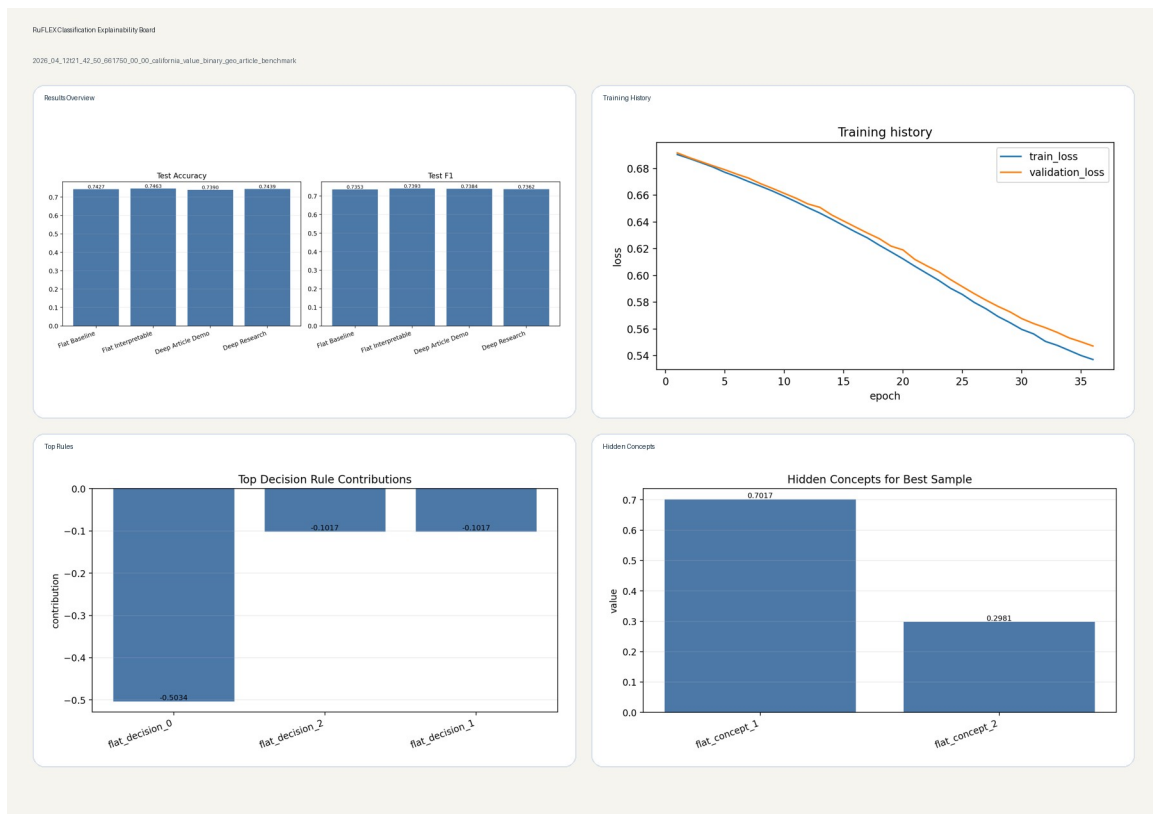


Fig. 4. Explainability board for the best classification configuration.

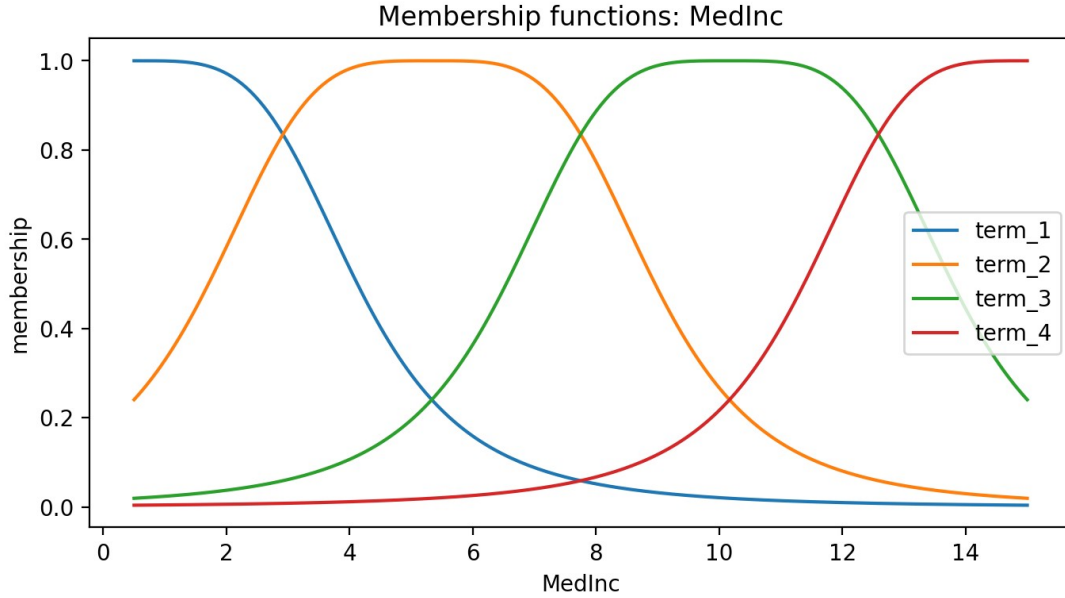


Fig. 5. Example membership functions used for interpretable fuzzification.

VI. Conclusion

RuFLEX combines interpretable rule-based mechanisms, differentiable fuzzy layers, deep fuzzy feature learning, a Python SDK, a visual workbench, and explainability tooling within one reproducible workflow. The current implementation is suitable for both regression and classification scenarios and provides a foundation for future work on spatial fuzzy modules, stronger logical traceability, and counterfactual analysis.

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